

Dynamic Min-Max Multi-Dimensional Reinforcement Backdoor Attacks and Orchestrated Closed-Loop Defense in Fairness-Aware Web Federated Finance

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Abstract

In the rapidly evolving web-based financial ecosystem where digital banking services become critical infrastructure for underserved communities, credit card fraud disproportionately affects vulnerable populations relying on financial platforms. However, previous studies overlook extreme data scarcity conditions, particularly at small-to-medium web banks that serve as crucial gateways for vulnerable communities. This paper addresses the fundamental challenge of building inclusive and secure financial systems operable at true web scale. To overcome this deficiency, we propose a novel web-based fairness-aware federated fraud detection model, CLARF, which utilizes the designed privacy-enhanced representation fusion and fraud-aware contrastive learning modules to enhance detection performance under conditions of data scarcity and label imbalance. Furthermore, current federated fraud detection systems critically neglect vulnerability to backdoor attacks, where malicious actors can implant hidden triggers during model aggregation, compromising system integrity. We propose a novel dynamic web Min-Max adversarial game framework where attackers employ hybrid multi-stage reinforcement learning with multi-dimensional reward mechanisms to dynamically evolve triggers that achieve excellent tradeoff between stealthiness and effectiveness. Defender adapts a closed-loop Selection-Evaluation-Suppression framework where high-reliability clients are selected via Fisher information to carry out reverse trigger engineering. Then clients' confidence scores are calculated as weights to minimize Attack Success Rate (ASR) during aggregation. Extensive experiments on six financial fraud datasets demonstrate the superiority of CLARF model and Min-Max adversarial game paradigm compared with multiple SOTA models.

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CCS Concepts

• **Social and professional topics** → **Spoofing attacks**; • **Security and privacy** → **Economics of security and privacy**.

Keywords

Web Finance, Societal Fairness, Inclusive Finance, Federated Learning, Reinforcement Learning

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1 Introduction

Credit card fraud poses a severe financial threat in the rapidly evolving web-based financial ecosystem, with annual losses exceeding 16 billion dollars. In recent studies [2, 3, 36, 39, 44, 48, 51], web-scale federated learning (FL) has emerged as a privacy-preserving alternative by enabling collaborative model training across distributed web finance institutions without sharing raw data. However, existing FL frameworks for fraud overlook critical challenges: (1) Extreme data scarcity at individual banks serving as gateways for underserved communities in the digital economy, and (2) Ultra rare fraud instances degrading representation learning. For many small-to-medium web banks especially in remote areas play a crucial role in extending financial inclusion to underserved web users.

While FL addresses critical data privacy concerns in financial fraud detection, backdoor threats are seriously understudied vulnerabilities in the finance contexts. In recent researches, [5] amplifies malicious updates before aggregation to forcibly implant backdoors in the global model. [26] employs combinatorial trigger generation across clients to reinforce global backdoor patterns against catastrophic forgetting. However, researches above employ statically preset poisoning ratios and fixed trigger patterns, fundamentally

constraining their capability to balance attack potency with operational stability while exhibiting critical deficiencies in adaptive responsiveness to dynamic defense mechanisms.

As backdoor threats evolve, defensive countermeasures have emerged to fortify FL systems. This creates an urgent need for backdoor resilient FL frameworks tailored to credit card systems' unique data asymmetry and security requirements. In recent works, Neuron Pruning [41] eliminates dormant neurons susceptible to trigger activation through post-training model compression; FL-guard [32] implements client-side validation to filter malicious updates using cross-verification. However, defenses above rely on static aggregation of local gradients, rendering them ineffective under information asymmetry where the set of malicious clients remains unknown and incapable of adapting to dynamic backdoor under web-based financial ecosystems, where robust security is essential to protect vulnerable users and maintain fairness in inclusive finance.

To address the dual challenges of extreme data scarcity at small financial institutions and catastrophic class imbalance overlooked, we propose **Contrastive Learning Accelerated initialization Representation Fusion federated paradigm (CLARF)**, a federated paradigm that innovates through two synergistic breakthroughs: (1) The fairness-oriented privacy-preserving representation fusion module overcomes data scarcity by means of enabling cross-client knowledge transfer across different banks with encrypted feature embeddings; (2) The fraud-aware contrastive learning module counteracts ultra-rare fraud instances in compressed latent spaces, amplifying web-scale minority-class signals through hard negative mining without representation collapses. This framework empowers small institutions to build robust fraud detection models, thereby protecting vulnerable communities and enhancing economic resilience and fairness in inclusive web-based financial systems.

To tackle the shortage of both current backdoor attacks and defenses, we propose a pioneering Min-Max adversarial game architecture specifically designed for financial security challenges. Our attack strategy coordinates malicious clients via lightweight hybrid phased reinforcement learning with multi-dimensional rewards, enabling dynamic trigger optimization against evolving defenses, while our defensive countermeasure implements a closed-loop selection-evaluation-suppression system featuring Fisher-based client screening, dynamic trigger inversion, and confidence-weighted aggregation, achieving great ASR without main task degradation under real-world web deployment constraints. Our adversarial framework is essential for safeguarding web-based financial systems and protecting underserved communities from the disproportionate impact of security breaches. In brief, our contributions can be summarized in the following perspectives:

- We propose CLARF, a novel financial federated model to simultaneously address extreme data sparsity and severe fraud instance scarcity in small institutions serving vulnerable populations. CLARF leverages privacy-preserving representation fusion, sharing encrypted transaction embeddings across distributed financial institutions. Furthermore, it incorporates fraud-aware contrastive learning module optimized for heterogeneity and an adaptively accelerated initialization module to adapt to transaction patterns common

in real-world finance operations. This approach bridges data gaps while advancing financial inclusion for underserved communities through secure web technologies.

- We pioneer a dynamic min-max game attack strategy for web-scale adversarial environments that orchestrates malicious clients through a lightweight hybrid-phase reinforcement learning mechanism with multi-dimensional rewards. This approach simultaneously optimizes trigger patterns and poisoning tactics in real-time, achieving unprecedented tradeoff between attack main task performance, concealment and great poisoning potency, establishing a new paradigm for adaptive adversarial research in federated systems.
- Additionally, we establish a closed-loop detection evaluation-suppression defense architecture comprising: (1) Fisher information based clean client screening to identify participants with great credibility; (2) Dynamic reverse-trigger confidence estimation for real-time threat assessment; and (3) Adaptive weighted aggregation countermeasures to neutralize poisoned updates. This tripartite system achieves dynamic adversarial suppression under information asymmetry, delivering a lightweight, highly adaptive security paradigm for federated learning, suppressing backdoor ASR to a low level while maintaining main-task accuracy.
- Extensive experimental evaluations on six real-world datasets demonstrate state-of-art performance of our method.

2 Related Work

2.1 Federated Learning

Federated learning [11, 25, 47] has emerged as a pivotal paradigm for privacy-preserving collaborative training, with foundational frameworks like FedAvg [28]; FedProx [24]; FedNova [38] and Scaffold [18]. Recent innovations include FedSol [20] for stability and FedSAM [22] for privacy-aware aggregation. Cross-round optimization techniques like FedCDA [37] further reduced divergence via gradient alignment, while sample-level drift mitigation enhanced robustness in heterogeneous environments. Despite these advances, existing methods exhibit limitations in extreme data scarcity regimes critical for web-scale financial inclusion, potentially exacerbating digital divides by excluding institutions serving vulnerable communities from web-based collaborative intelligence, thereby hindering progress toward an equitable web financial ecosystem.

2.2 Backdoor Attacks against Federated Learning

Federated backdoor attacks [30, 42] have progressed from basic poisoning to advanced aggregation evasion strategies. Early work includes Sybil [14]; LIE [6]; DBA [44] and Cerp [27]. Recent innovations focus on persistent backdoors like [26] that survive model refinement and data-free attacks [23] that requires no local malicious data. Personalized FL vulnerabilities [12] exploit client-specific adaptations, while label-free attacks like [35], compromise vertical FL through feature space manipulation. These advances highlight critical security gaps in web-based financial ecosystems where attacks can disproportionately harm vulnerable populations relying

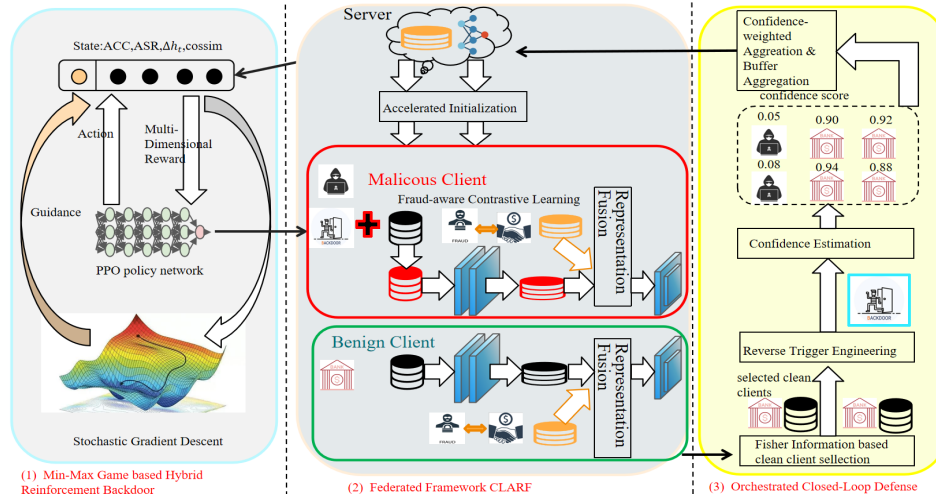


Figure 1: Overview of the proposed CLARF model: (1). Clients process transactions using inclusive privacy-preserving representation fusion contrastive learning and accelerated initialization to overcome data scarcity; (2). Malicious clients employ multi-stage RL with multi-dimensional rewards to dynamically optimize effective yet stealthy triggers; (3). Defender performs dynamic backdoor threat mitigation through Fisher-based screening, trigger inversion, and confidence-weighted aggregation.

on digital banking services, underscoring the need for adaptive protection mechanisms in inclusive Web finance.

2.3 Defense against Federated Backdoor

Defenses against federated backdoor threats [10, 15, 40, 50] have evolved from Byzantine-robust aggregation to adaptive trigger detection frameworks. Foundational work includes Mkrum [7]; Median [49]; Bulyan [29]; FLTrust [8] and FLAME [31]. Recent approaches leverage parameter disparity dissection [16] to identify backdoor-critical layers and bidirectional election mechanisms [34] for client validation. Advanced detection techniques include reconstruction error analysis FedReDefense [45] and gradient direction alignment inspection [46], achieving up to satisfying attack identification accuracy. However, these methods exhibit critical limitations against adaptive multi-client attacks like SADBA [13] that dynamically evade static detection signatures in real-world web financial systems, failing to provide protection to vulnerable populations under dynamic backdoor.

2.4 Web Federated Finance and Social Good

In the domain of web finance, FL has been leveraged to enhance financial risk management and service transparency for inclusive digital economies. For instance, [17] proposed FedRisk, a knowledge graph enhanced FL framework for efficient risk prediction across web-based financial institutions. [33] and [53] are utilized in fraud detection. Similarly, [4] integrated explainable AI with FL to improve transparency in online banking models, crucial for user trust in web financial services. In social good contexts, [40] combined FL with blockchain in medical IoT for disease diagnosis, enhancing data security in web-enabled healthcare systems. [9] demonstrated FL’s role in maximizing social welfare via public goods game models applicable to web-based cooperative platforms. Additionally, [19] surveyed privacy-preserving FL strategies in e-commerce and

finance, highlighting potential to address data heterogeneity in web-scale applications. While these advances underscore FL’s value for social good, they fall short of explicitly safeguarding vulnerable populations from financial fraud in modern web ecosystems, indicating a critical need for more inclusive security frameworks ensuring web economic fairness.

3 Federated Framework CLARF

The CLARF model includes three core modules: Fairness-Oriented Privacy-Enhanced Representation Fusion Module, Fraud-Aware Contrastive Learning Module and Adaptively accelerated Initialization Module.

3.1 Fairness-Oriented Privacy-Enhanced Representation Fusion Module

We empower small financial institutions which provide service for underserved communities and suffer from data scarcity by introducing a privacy-preserving cross-client knowledge distillation architecture, augmented with a feature representation cache to facilitate efficient knowledge transfer. This architecture stores intermediate feature embeddings from designated model layers rather than raw data, ensuring compliance with financial privacy regulations. During each federated round, we dynamically update the cache using stratified sampling which maintains a fixed normal-to-fraud ratio to guarantee sufficient minority class representation. Subsequently, we generate diverse synthetic instances through optimized feature fusion, implementing linear interpolation as shown below:

$$\tilde{z}_i \leftarrow \alpha z_i + (1 - \alpha) z_i^{cache}, \alpha \in (0, 1), \quad (1)$$

where z_i denotes the feature-label pair in local dataset, z_i^{cache} represents the one in the cache and $\tilde{z}_i$ stands for the synthetic feature-label pair generated through representation fusion. We incorporate a privacy preservation mechanism through Feature Privacy Loss

L_{fp} which minimizes the distance covariance between intermediate feature embeddings z_i and the raw data x_i to mitigate raw attribute leakage. Specifically,

$$L_{fp} = \left(\frac{D^2(z_i, x_i)}{D(z_i, z_i)D(x_i, x_i)} \right)^{1/2}, \quad (2)$$

where D denotes squared distance covariance. Optimizing this loss function reduces mutual information while preserving utility. This approach ensures that shared embeddings contain minimal raw data information to prevent privacy leakage, especially for the underserved communities. By overcoming data scarcity through privacy-preserving techniques, we actively bridge the digital divide in the financial sector, enabling more fair participation in the inclusive web finance ecosystem.

3.2 Fraud-Aware Contrastive Learning Module

Conventional federated contrastive learning addresses statistical heterogeneity and class imbalance by aligning feature distributions across clients. The contrastive loss function is formally defined as:

$$\mathcal{L}_{CL} = \sum_{i \neq j, y_i = y_j} -\log \frac{e^{\langle \phi_i, \phi_j \rangle / \tau}}{\sum_k e^{\langle \phi_i, \phi_k \rangle / \tau}}, \quad (3)$$

where τ is temperature parameter, ϕ_i is the feature representation of data x_i and y_i denotes the label of data x_i . In order to prevent representation collapse under extreme class imbalance, we add punishment on pairwise same class instances with exceedingly close feature representation. We formulate this punishment as follows:

$$\mathcal{L}'_{CL} = \mathcal{L}_{CL} + \sum_{\|\phi_i - \phi_j\|^2 < d} \log e^{\langle \phi_i, \phi_k \rangle / \tau}. \quad (4)$$

This module solves the problem of data heterogeneity without representation collapse which is common in contrastive learning.

3.3 Adaptively Accelerated Initialization Module

To prevent global model divergence caused by inter-client inconsistency, we initialize local models each round with both global parameters(θ^g) and global gradient approximations($\tilde{g}^g$). Since exact global gradients $\nabla \mathcal{L}(\theta^g; \mathcal{D})$ are incomputable without client data access $\mathcal{D}$, we derive $\tilde{g}^g$ from the momentum of global parameter updates and initialize local models.

$$\tilde{g}_t^g = \beta \cdot \Delta \theta_t + (1 - \beta) \cdot \tilde{g}_{t-1}^g, \quad (5)$$

$$\theta_i^t \leftarrow \theta_t^g + \lambda \cdot \tilde{g}_t^g, \quad (6)$$

where $\Delta \theta_t = \theta_t - \theta_{t-1}$ represents the global update vector, β denotes momentum decay coefficient, and λ controls acceleration speed.

4 Min-Max Game based Hybrid Reinforcement Backdoor

This section contains three core components: Min-Max Objective Function, Multi-Dimensional Reward Reinforcement Learning and Hybrid Multi-Stage Training Strategy.

4.1 Min-Max Objective Function

In the attack-defense game of federated learning backdoor attack, the attacker and the defender form a min-max optimization problem. Attackers aim to minimize attack detectability and maximize attack success rates, while defenders maximize detection of malicious models through robust aggregation rules. The overall objective function is defined as follows:

$$\min_{\delta_{re}} \max_{R^t, \delta} \left(\sum_{i \in \mathcal{S}_a} p_i^t - \sum_{j \in \mathcal{S} \setminus \mathcal{S}_a} p_j^t + \lambda \sum_{i \in \mathcal{S}_a} r_i^t + \vec{\lambda}' \cdot \vec{U} \right), \quad (7)$$

$$\vec{U} = (ASR_t, ACC_t, \Delta h_t, \text{cossim}_t), \quad (8)$$

where the objective function integrates four strategic components, i.e., ASR_t , ACC_t , Δh_t , cossim_t . The first two terms denotes disguise scores, which quantify evasion capabilities against detection systems. The third term regulates backdoor intensity through trigger magnitude control, while the final term optimizes the critical tradeoff between attack effectiveness (i.e., ASR) and stealthiness. Formally, for client i in round t : $p_i^t \in [0, 1]$ represents the confidence score, $r_i^t \in [0, 1]$ denotes backdoor injection ratio, with hyperparameters λ and $\vec{\lambda}'$ balancing attack objectives. The set of all participating clients is noted as $\mathcal{S}$, the subset of malicious clients is noted as $\mathcal{S}_a \subseteq \mathcal{S}$ and $\mathcal{S} - \mathcal{S}_a$ represents the subset of benign clients, enabling precise adversarial control in federated environments. We define a performance vector $\vec{U}$ quantifying four critical attack dimensions: 1). Attack Success Rate (ASR); 2). Model Accuracy (ACC); 3). Malicious Model Deviation ($\Delta h^t = \|h^{mal,t} - h^{nor,t}\|_F$); and 4). Intra-Attacker Cosine Similarity (cossim_t). The cossim_t is defined as follows:

$$\text{cossim}_t = \sum_{i \neq j} \frac{\langle h_i^t, h_j^t \rangle}{\|h_i^t\| \cdot \|h_j^t\|}. \quad (9)$$

Malicious clients collaboratively optimize the poison ratio vector $R_t = (r_1^t, r_2^t, \dots, r_{\mathcal{S}_a}^t)$ and backdoor trigger δ to maximize the game objective function through hybrid multi-stage reinforcement learning with multi-dimensional reward mechanism. In contrast, the defender minimizes game objective function via backdoor trigger reconstruction δ_{re} . This adversarial dynamic threatens the integrity of web finance, where successful attacks may undermine trust in digital banking services and disproportionately harm vulnerable low-income individuals who rely on inclusive participation. Thus, advancing min-max game theory in this context is crucial for safeguarding the whole web economy.

4.2 Multi-Dimensional Reward Reinforcement Learning

We formalize backdoor trigger optimization as a Markov Decision Process (MDP) to enable adaptive injection of stealthy yet effective triggers into global federated models. The MDP is defined by:

- State space $s_t: s_t = (ASR_t, ACC_t, \Delta h_t, \text{cossim}_t)$, which is mentioned in the last subsection.
- Action space a_t : Directional perturbation Δx to adjust trigger δ per dimension. The discrete action space is adopted here to reduce computational cost.

- Reward r_t : $r_t = U(ASR_t, ACC_t, \Delta h_t, cossim_t)$, where utility function $U(\cdot)$ is a non-decreasing neural network that approximates Pareto-optimal tradeoffs between stealth and effectiveness.

4.3 Hybrid Multi-Stage Training Strategy

To overcome reinforcement learning's cold start problem in backdoor optimization, we design a hybrid multi-stage training strategy partitioning federated rounds into three phases:

- Freezing Stage: Initialize triggers Δx via gradient descent:

$$\Delta x \leftarrow \Delta x - \alpha \nabla_{\Delta x} \sum_{i \in S} \ell_{h_i}(x_i + \Delta x, \hat{y}_i), \quad (10)$$

where RL policy does neither inference nor training due to RL's poor performance in this stage.

- Warm-Up Stage: Triggers are optimized by gradient descent while pretraining the RL policy network $\pi_\theta(a|s)$ through behavioral cloning in gradient-generated actions.
- We deploy constrained Proximal Policy Optimization (PPO) with KL-divergence regularization, which enables dynamic trigger evolution while maintaining policy stability, as shown below:

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{s \sim \rho_\pi} \left[\frac{\pi_\theta(a|s)}{\pi_{\theta_{old}}(a|s)} r_t - \beta \text{KL}(\pi_{\theta_{old}} \parallel \pi_\theta) \right]. \quad (11)$$

5 Equality-Preserving Orchestrated Closed-Loop Defense

This section introduces an equality-preserving closed-loop defense framework specifically designed to counteract dynamic backdoor threats while maintaining fraud detection performance. The proposed architecture comprises three core modules that operate in a coordinated sequential pipeline including Fisher information based clean client screening, dynamic reverse trigger confidence estimation and confidence-weighted aggregation countermeasure.

5.1 Fisher Information Based Clean Client Screening

We utilize fisher information to identify benign clients $\mathcal{K}_{clean} \subseteq S$ without prior knowledge of malicious set $\mathcal{S}_a$, leveraging parameter importance to amplify malicious anomalies. Fisher Information is adapted to quantify sensitivity of parameters to data distribution shifts and is critical for detecting deviations induced by backdoors. The definition of Fisher Information Matrix (FIM) is defined as follows:

$$F_k = \frac{1}{|D_k|} \sum_{(x,y) \in D_k} (\nabla_w \log p(y|x, w_k))^2, \quad (12)$$

where F_k and w_k denote fisher information and local model parameters of client k respectively. This formulation prioritizes parameters most susceptible to backdoor triggers, eliminating bias against all clients, which is a key aspect of equality-preserving design. The central server aggregates local gradients g_k^t from participating clients, and computes importance-weighted gradients $\tilde{g}_k^t = g_k^t \odot F_k$, thereby amplifying deviations in trigger-sensitive parameters while suppressing non-critical updates. Through federated weighted averaging aggregation, we derive a global importance-weighted gradient

$\tilde{g}^t = \sum_k \omega_k \tilde{g}_k^t$. Finally, we identify the clean client set $\mathcal{K}_{clean}$ by selecting the top clients with a minimum value of $\|\tilde{g}^t - \tilde{g}_k^t\|_2$, ensuring statistical alignment with the trusted gradient distribution. This step reinforces equality by basing exclusion solely on anomalous patterns rather than inherent client characteristics to carry out following defense procedures, achieving secure and equitable collaborations for all financial institutions in real-world finance.

5.2 Dynamic Reverse Trigger Confidence Estimation

This module exemplifies the equality-preserving design by ensuring that client trustworthiness is assessed impartially without bias, aligning with the goal of fair and secure collaborations. To quantify client trustworthiness amid unknown malicious set $\mathcal{S}_a$, we construct an auxiliary global model Θ_a^t that intentionally amplifies backdoor susceptibility through uniform update aggregation:

$$\Theta_a^t = \frac{1}{|S|} \sum_{i \in S} \Theta_i^t, \quad (13)$$

where Θ_i^t denotes the local model of client i . The uniformly aggregated auxiliary model Θ_a^t exhibits intentional susceptibility amplification, systematically misclassifying triggered instances into the attacker-specified target label.

The proposed federated compliant inverse engineering method, guided by auxiliary model Θ_a^t , successfully reconstructs the optimal backdoor trigger δ^* while evading detection in cross-client gradient aggregation scenarios, as shown below:

$$\delta_k^* = \arg \min_{\delta} \mathbb{E}_{(x,y) \sim \mathcal{D}_{clean}} [\ell_{CE}(\Theta_a^t(x + \delta), \hat{y})], \quad (14)$$

$$\delta^* = \sum_{k \in \mathcal{K}_{clean}} \omega_k \delta_k^*, \quad (15)$$

where $\mathcal{D}_k$ denotes aggregated datasets from Fisher-screened high-credibility clients. To preserve data privacy, each trusted client $k \in \mathcal{K}_{clean}$ computes local δ_k^* via on-device optimization. Then central server derives global trigger through privacy-preserving aggregation. The privacy-preserving nature of this on-device optimization safeguards client data confidentiality, a cornerstone of equitable federated learning.

We compute client confidence scores through a federated diagnostic protocol with two steps:

- Trigger susceptibility assessment: For each client i , measure susceptibility to the reconstructed trigger δ^* across trusted datasets $\mathcal{D}_k$ from clean clients $k \in \mathcal{K}_{clean}$:

$$C_{i,k}^t = \sum_{x \in \mathcal{D}_k} \mathbb{I}(\Theta_i^t(x \oplus \delta^*) = \hat{y}), k \in \mathcal{K}_{clean}, \quad (16)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function, Θ_i^t is client i 's local model, and $\hat{y}$ is the attacker-specified target label.

- Robustness-aware confidence scoring: Compute global susceptibility via weighted aggregation and derive confidence as:

$$p_i^t = 1 - \sum_{k \in \mathcal{K}_{clean}} \frac{|\mathcal{D}_k|}{\sum_{k \in \mathcal{K}_{clean}} |\mathcal{D}_k|} C_{i,k}^t. \quad (17)$$

5.3 Dynamic Confidence Weighted Aggregation Countermeasure

As the final phase of the equality-preserving closed-loop defense, this aggregation mechanism dynamically adjusts client contributions based on rigorously derived confidence scores, preventing malicious influence while preserving the integrity of benign updates. We utilize confidence scores to perform reweighted aggregation, suppressing malicious updates while preserving model integrity on benign data. This constitutes the final phase of our selection-evaluation-suppression defense pipeline.

$$\Theta^{t+1} = \Theta^t + \eta \frac{\sum_{i \in K} p_i^t g_i^t}{\sum_{i \in K} p_i^t}. \quad (18)$$

The confidence-weighted aggregation constitutes the suppression phase of our defense, dynamically reweighting client updates based on trust metrics derived from Fisher screening and trigger inversion diagnostics. Specifically, each client's contribution is scaled by its confidence score $p_i^t \in [0, 1]$. This formulation ensures malicious suppression while preserving benign functionality. By scaling updates with confidence scores, the system ensures that all clients are treated proportionally to their trustworthiness, fostering a collaborative environment that prioritizes security and fairness.

6 Experiments

6.1 Experimental Setup

Datasets. To evaluate the effectiveness of our model in real-world financial risk detection scenarios, we conduct experiments on six financial transaction datasets, including three public datasets and three proprietary datasets. The statistical characteristics of all datasets are summarized in Table 1.

Private Datasets. Our evaluation has been conducted on three proprietary datasets (Private-1, Private-2, Private-3) provided by major Chinese commercial banks, comprising a total of 2,601,288 transaction records. These datasets present substantial class imbalance, with fraud ratios of 15.62%, 1.81%, and 0.65%, respectively, thereby reflecting diverse operational risk profiles encountered in real financial systems. We construct high-dimensional risk representations from raw banking transaction logs, covering four major feature groups including transaction context, payment channel constraint, entity interaction history as well as device and IP consistency group. Transaction context includes temporal attributes, transaction amount and merchant identifiers. Payment channel constraint includes cumulative electronic payment limits and per-transaction caps. Entity interaction history incorporates receiver identifiers and historical transaction outcomes. Device and IP consistency group includes geographic region indicators, IP reuse signals and mobile device matching status. This multi-view feature design enables fine-grained fraud detection across transactions spanning various provincial administrative regions.

Public Datasets. The C_1 dataset from [43] is a simulated financial dataset designed for semi-supervised fraud detection research. The C_2 dataset, introduced by [52], focuses on predicting credit card default risk among Taiwanese cardholders. The C_3 dataset consists of European card transactions, including 30 input features, among which 28 are PCA-transformed to preserve confidentiality.

By combining proprietary banking data with widely used public benchmarks, our evaluation covers a broad spectrum of transaction behaviors and risk patterns, enabling a robust assessment of model generalization under varying levels of fraud prevalence, feature abstraction, and class imbalance.

Dataset	Record	Fraud	Legitimate	Unlabeled
C_1	77881	5256	24387	48238
C_2	30000	6636	23364	0
C_3	284807	492	284315	0
Private-1	666592	104112	562480	0
Private-2	691661	12507	679154	0
Private-3	1243035	8086	1234949	0

Table 1: Statistics of six datasets.

Baselines. We conduct comprehensive benchmarking against 11 SOTA federated learning framework, 6 SOTA federated backdoor approaches and 8 SOTA federated defense approaches to demonstrate the superiority of our proposed CLARF model and our backdoor/defense approach. 11 federated frameworks include FedAvg [28], FedProx [24], FedNova [38], Scaffold [18], FedDyn [1], FedNTD [21], FedSOL [20], FedCDA [37] and FedSam [22]. 6 attack approaches includes Sybil [14], LIE [6], DBA [44], FCBA [26], LP Attack [54] and DarkFed [23]. 8 defense approaches includes MKrum [7], Median [49], Bulyan [29], FLAME [31], FLTrust [8], FDCR [16], FedREDefense [45] and Snowball [34].

Evaluation Metrics and Hyperparameter Settings. In the experiment, we utilize AUC, F1, average precision (AP), attack success rate (ASR) to evaluate the effectiveness of our model. We configure the federated learning system with 50 global communication rounds, and α is set to 0.3 to mimic data heterogeneity across clients. For federated framework evaluation, we deploy 200 clients with 8% participation rate per round, while backdoor attack and defense experiments utilize 100 clients with 8% participation rate and 25% malicious client ratio.

6.2 Fraud Detection Performance

Our comparative evaluation against 11 state-of-the-art federated learning frameworks in Table 2 demonstrates CLARF's superior fraud detection capabilities across all six datasets. CLARF consistently outperforms existing approaches, achieving the highest scores in AP (0.9417, 0.9736, 0.9790) and F1 (0.9482, 0.9568, 0.9385) across Private-1, Private-2, and Private-3 datasets. Remarkably, CLARF improves AP by 4.20% over the strongest baseline (FedProx) on Private-1 and attains 7.12% and 8.01% AP gains on Private-2 and Private-3, which validates our framework's robust handling of data heterogeneity and adversarial conditions while maintaining competitive AUC performance within 1% of the top baseline.

6.3 Backdoor Attack/Defense Performance

Our experimental analysis in Table 3 and Figure 2 reveals two critical phenomena: First, our backdoor attack module achieves perfect 100% ASR against 6 SOTA defense approaches because the RL attack continuously adapts trigger to the fixed aggregation and filtering mechanisms of existing defenses, eventually converging to

Table 2: Fraud detection performance on six datasets compared with 11 SOTA models.

Method	Private-1			Private-2			Private-3			C ₁			C ₂			C ₃		
	AP	F1	AUC	AP	F1	AUC	AP	F1	AUC	AP	F1	AUC	AP	F1	AUC	AP	F1	AUC
FedAvg	0.8956	0.9012	0.9854	0.8915	0.9012	0.9845	0.8945	0.9031	0.9854	0.8170	0.8573	0.9448	0.8159	0.8418	0.9149	0.9408	0.9421	0.9996
FedNova	0.8843	0.8926	0.9879	0.8841	0.8956	0.9882	0.8855	0.8986	0.9879	0.7902	0.8363	0.9328	0.8111	0.8388	0.9139	0.9421	0.9464	0.9999
FedProx	0.9037	0.9107	0.9902	0.9089	0.9178	0.9904	0.9064	0.9203	0.9921	0.8161	0.8649	0.9425	0.8113	0.8380	0.9147	0.9453	0.9477	0.9999
SCAFFOLD	0.8975	0.9021	0.9846	0.8951	0.9064	0.9848	0.8966	0.9064	0.9836	0.7027	0.6704	0.8074	0.6797	0.6765	0.7973	0.9397	0.9411	0.9998
FedDyn	0.7202	0.7346	0.9608	0.7239	0.7473	0.9610	0.7272	0.7407	0.9604	0.7818	0.8625	0.9446	0.7795	0.8232	0.8955	0.7993	0.7933	0.8325
FedNTD	0.8883	0.7204	0.7212	0.8834	0.8909	0.9835	0.8880	0.8896	0.9838	0.7428	0.8419	0.9417	0.8038	0.8249	0.9080	0.9304	0.9312	0.9997
FedSam	0.8707	0.8810	0.9924	0.8765	0.8841	0.9917	0.8726	0.8897	0.9921	0.8989	0.9149	0.9677	0.5019	0.8841	0.6673	0.9228	0.9279	0.9844
FedSol-fixed	0.9014	0.9096	0.9870	0.9014	0.9140	0.9870	0.9051	0.9175	0.9875	0.7373	0.7125	0.8813	0.6920	0.6784	0.8096	0.9280	0.9321	0.9996
FedSol-adaptive	0.8901	0.8964	0.9856	0.8874	0.8957	0.9857	0.8925	0.8978	0.9859	0.8117	0.8569	0.9424	0.8091	0.8358	0.9138	0.9242	0.9265	0.9997
FedBSS	0.8967	0.9023	0.9841	0.8962	0.9043	0.9847	0.8948	0.9068	0.9844	0.8160	0.7745	0.8984	0.8710	0.8980	0.9712	0.9481	0.9490	0.9993
FedCDA	0.8746	0.8795	0.9864	0.8764	0.8743	0.9868	0.8788	0.8758	0.9864	0.9482	0.9488	0.9493	0.8534	0.8725	0.9673	0.9460	0.9430	0.9964
CLARF	0.9417	0.9482	0.9877	0.9736	0.9568	0.9946	0.9790	0.9385	0.9923	0.9532	0.9584	0.9663	0.9318	0.9116	0.9810	0.9612	0.9672	0.9946

Table 3: Federated backdoor approach performance compared with 6 SOTA backdoor approaches. RED denotes FedREDefense.

	Sybil	DBA	LIE	FCBA	LP	DarkFed	Our
MKrum	0.3130	0.1342	0.8462	0.7983	0.9518	1.0	1.0
Bulyan	0.1333	0.1783	0.7659	0.8291	0.9624	1.0	1.0
Median	0.1482	0.1999	0.5653	0.7884	0.9351	1.0	1.0
FLAME	0.3669	0.1296	0.2496	0.6294	0.8846	0.9754	1.0
FLTrust	0.2617	0.1572	0.2840	0.5716	0.8154	0.9267	1.0
RED	0.0	0.0	0.0	0.4068	0.5975	0.6239	0.6672
FDCR	0.1098	0.0	0.0387	0.4623	0.5583	0.7560	1.0
Snowball	0.2597	0.0975	0.2780	0.5676	0.7954	0.5047	0.8060

Table 4: Ablation Study of Federated Backdoor on ASR (DTO stands for Dynamic Trigger Optimization).

	w/o DTO	Our Backdoor
MKrum	0.4182	1.0
Bulyan	0.6250	1.0
Median	0.3887	1.0
FLAME	0.4019	1.0
FLTrust	0.3030	1.0
FedREDefense	0.0	0.6672
FDCR	0.2384	1.0
Snowball	0.1690	0.8060

trigger configurations that consistently evade their static rejection criteria. Second, our defense module achieves perfect 100% ASR detection against our own advanced backdoor attack, outperforming all tested SOTA defenses because the defense proposed effectively constrains malicious updates before aggregation, enabling confidence weighted aggregation to completely neutralize the proposed backdoor without interfering with benign optimization.

6.4 Ablation Study

We conduct comprehensive ablation studies in both attack and defense frameworks. For the backdoor attack methodology, we construct a variant without dynamic trigger optimization. On the defense side, we systematically disable components of our orchestrated framework either the clean client selection is disabled or the reverse trigger engineering is disabled or both. In Table 4, the dynamic trigger optimization module proves indispensable. Without it, attack variants maintain moderate ASR protection, but with it, our attack achieve 100% ASR against most defenses, demonstrating how dynamically evolved triggers. For defenses in Table 5, clean client selection alone reduce ASR to 79.76%, while standalone reverse

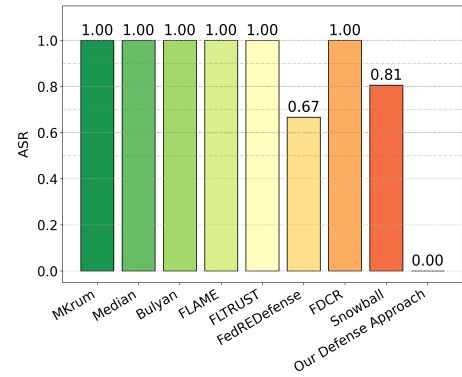


Figure 2: Federated backdoor defense approach performance compared with 8 SOTA defense approaches.

trigger gets 100% ASR, but their combination achieves 0% ASR. This synergy works because clean client screening first mitigates malicious clients' influence via Fisher-information analysis, enabling the reverse engineering module to reconstruct residual dynamic triggers from the sanitized update pool with great accuracy.

Additionally, we perform an ablation experiment on the CLARF module. Table 6 shows that the full configuration with all components enabled achieves the highest performance (AP=0.9532, F1=0.9584), while disabling any single component leads to a noticeable drop. For example, without Fusion, AP decreases to 0.9507, and without Contrastive learning, AP reduces to 0.9458. This synergy highlights the necessity of integrating all components for optimal model robustness in handling data scarcity.

6.5 Sensitivity Analysis

Our comprehensive hyperparameter sensitivity analysis demonstrates the robustness of the proposed defense framework against adversarial attacks and the stability of the CLARF framework across critical operational parameters. As evidenced in Figure 3, ASR maintains zero values across the entire spectrum of Heterogeneity Degree (0.5-1.0) and Malicious Client Ratio (0.15-0.4), confirming our defense mechanism's resilience against distributional shifts and adversarial infiltration attempts. For the CLARF framework: AUC

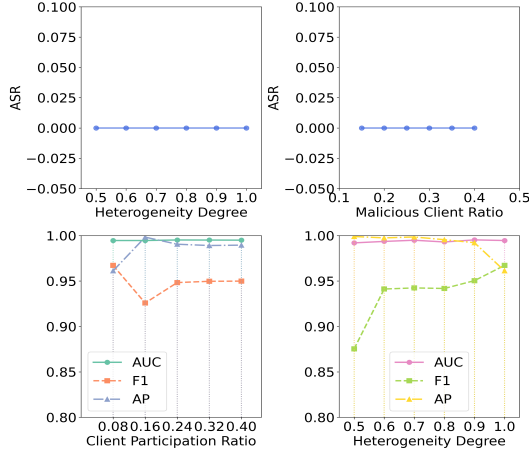


Figure 3: Sensitivity analysis for orchestrated defense on proposed attack (upper two) and CLARF (lower two).

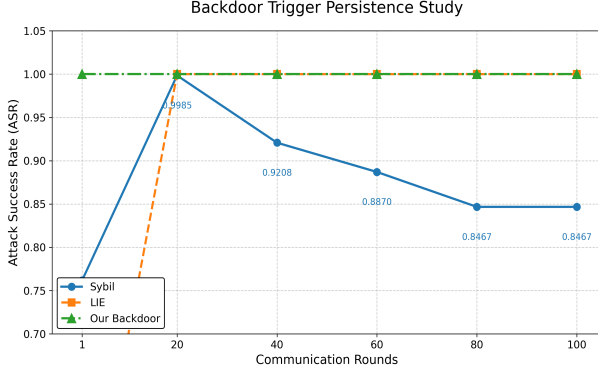


Figure 4: Backdoor Trigger Persistence Analysis

maintains high across Client Participation Ratios (0.08-0.4) and Heterogeneity Degrees (0.5-1.0), with AP consistently sustains superior despite parameter fluctuations, validating CLARF’s operational robustness.

6.6 Backdoor Trigger Persistence Study

Table 5: Ablation Study of Our Defense. (‘✓’ denotes chosen, ‘-’ means not chosen, ‘RTE’ means Reverse Trigger Engineering).

Clean Client Selection	RTE	ASR
-	-	1.0
-	✓	1.0
✓	-	0.7976
✓	✓	0.0

This persistence study reveals backdoor robustness post-adversary departure (round 20+) in Figure 4. In contrast, Sybil exhibits progressive degradation, declining from 99.85% at round 20 to 84.6% by round 100, demonstrating that Sybil’s gradient poisoning remains vulnerable to benign update erosion, as federated averaging gradually dilutes its transient perturbations. The sustained 100% ASR for

Table 6: Ablation Study of CLARF framework. (‘✓’ and ‘-’ denotes chosen and not chosen respectively).

Fusion	Contrastive	Init	AP	F_1
-	-	-	0.9029	0.9119
✓	-	-	0.9172	0.9181
-	✓	-	0.9251	0.9243
-	-	✓	0.9260	0.9319
-	✓	✓	0.9507	0.9525
✓	-	✓	0.9458	0.9434
✓	✓	-	0.9420	0.9431
✓	✓	✓	0.9532	0.9584

LIE and our backdoor confirms their catastrophic forgetting resistance, which is a critical property absent in conventional attacks against federated unlearning mechanisms.

6.7 Discussion

Our research fundamentally redefines federated backdoor security for financial ecosystems through a holistic min-max adversarial paradigm that addresses critical gaps in existing approaches. We achieve consistent cross-domain aggression bypassing pitfalls of data heterogeneity. Our adaptive hybrid mechanism leverages reinforcement learning to maintain potency, which is designed strictly for defensive stress-testing in inclusive finance not for malicious exploitation. We emphasize that this framework must not be employed for any malicious purposes beyond controlled security auditing. Defensively, we utilize an integrated shield combining Fisher-based client screening, dynamic trigger inversion, and confidence weighted aggregation.

7 Conclusion

This study presents CLARF, a unified federated framework that significantly improves performance under extreme data scarcity through a fairness-oriented privacy-preserving representation fusion module and fraud-aware contrastive learning mechanism. CLARF ensures equitable access to secure financial services for all users, including vulnerable populations, by mitigating biases in digital ecosystems. For security, we pioneer a min-max adversarial paradigm which employs hybrid reinforcement learning with multi-dimensional rewards to dynamically evolve triggers, achieving stealth-persistence tradeoffs, and our defense establishes a closed-loop Fisher trigger defense system integrating clean-client screening via Fisher information, reverse-trigger engineering, and confidence-weighted aggregation, upholding fairness by protecting participants from diverse backgrounds to achieve safer and more justified finance environment. Extensive experiments on six financial datasets reveals that our CLARF model and Min-Max adversarial game paradigm significantly outperform multiple SOTA models. Future work will explore real-time regulatory frameworks for the underserved communities in inclusive finance.

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A Baseline

We conduct comprehensive benchmarking against 11 state-of-art federated learning frameworks to demonstrate the superiority of our proposed CLARF framework on 6 financial datasets. FedAvg [28] represents basic federated optimization frameworks, while FedProx [24], FedNova [38], Scaffold [18], FedDyn [1], FedNTD [21], FedSOL [20], FedCDA [37], FedSam [22], and FedBSS [46] constitute advanced federated frameworks addressing non-IID data challenges through heterogeneity handling, gradient correction, and knowledge distillation techniques. We validate the superiority of our

Algorithm 1 Pseudo Code of Our CLARF Model

Require: Global model Θ^t , gradient approximation $\tilde{g}^t$, clients $\{C_i\}$, cache ratio ρ , temperature τ

Ensure: Updated global model Θ^{t+1}

- 1: **Server executes:**
- 2: Distribute $(\Theta^t, \tilde{g}^t)$ to selected clients S_t ;
- 3: **Client i executes in parallel:**
- 4: $\theta_i^0 \leftarrow \Theta^t + \lambda \tilde{g}^t$; ▷ Warm-up initialization
- 5: $C_i \leftarrow \text{UpdateCache}(\mathcal{D}_i, \rho)$;
- 6: **For each local epoch:**
- 7: Sample batch $\mathcal{B}$ from $\mathcal{D}_i$;
- 8: $\tilde{z} \leftarrow \alpha z + (1 - \alpha)z_{\text{cache}}$; ▷ Feature fusion
- 9: Compute contrastive loss $\mathcal{L}_{CL}$ with collapse penalty;
- 10: $\mathcal{L}_{CE} \leftarrow \text{CrossEntropy}(f_{\theta}(\tilde{z}), y)$;
- 11: Update θ_i via gradient descent on $\mathcal{L}_{total}$;
- 12: **Server aggregates:**
- 13: $\Theta^{t+1} \leftarrow \Theta^t + \frac{1}{|S_t|} \sum \Delta \theta_i^t$;
- 14: $\tilde{g}^{t+1} \leftarrow \beta \Delta \Theta + (1 - \beta)\tilde{g}^t$. ▷ Momentum update

novel backdoor attack strategy and orchestrated defense framework through comprehensive benchmarking against 6 SOTA federated backdoor attacks and 8 SOTA federated defense approaches to achieve fairness in inclusive finance, establishing new performance standards in adversarial robustness. Sybil [14] and LIE [6] represent traditional model poisoning attacks, while DBA [44], FCBA [26], LP Attack [54], and DarkFed [23] constitute advanced stealthy backdoor attacks employing collusion, persistence, and data-free strategies. MKrum [7], Median [49], and Bulyan [29] represent classical Byzantine-robust aggregation defenses, while FLAME [31], FLTrust [8], FDCR [16], FedREDefense [45], and Snowball [34] constitute advanced backdoor-specific defense frameworks employing trust bootstrapping, parameter disparity analysis, and adversarial election mechanisms.

B Pseudo Code of CLARF

Our CLARF framework (Algorithm 1) implements a privacy-preserving federated learning paradigm specifically designed for financial ecosystems, addressing critical challenges of data scarcity and heterogeneity in cross-institutional collaboration. The algorithm initiates with federated model initialization (Line 2) where the server distributes global parameters Θ^t and gradient approximations $\tilde{g}^t$ to participating clients, enabling accelerated convergence in financial environments. Each client performs local model warm-up (Line 4) using both global parameters and momentum-based gradients, significantly reducing divergence in heterogeneous data distributions. The core innovation lies in the privacy-enhanced representation fusion module (Lines 5-8), where clients maintain encrypted feature caches. Through stratified sampling with fixed fraud-to-normal ratios, the cache preserves minority class representations critical for financial inclusion in underserved communities. The module generates synthetic instances via linear interpolation between local and cached features (Line 8), enhancing data diversity while maintaining GDPR compliance through feature-level differential privacy. The optimized contrastive loss $\mathcal{L}_{CL}$ (Line 9) incorporates collapse prevention penalties to maintain representation stability under web-scale data heterogeneity. Simultaneously, the classification module computes cross-entropy loss on synthetic samples (Line 10), balancing representation learning and task performance for real-time fraud detection in web inclusive finance. Local model updates (Line 11) employ adaptive gradient descent optimized for edge device constraints, while the server aggregates updates through federated averaging (Line 13) enhanced with momentum-based acceleration (Line 14) for improved stability in inclusive finance. This integrated approach demonstrates effective protection of vulnerable populations in digital banking ecosystems.

C Pseudo Code of Our Backdoor Approach

Our federated backdoor attack algorithm (Algorithm 2) employs a multi-objective optimization strategy to dynamically balance attack potency and stealthiness under web financial situations. The algorithm initiates by reserving a subset of local data $\mathcal{D}_i^{rev}$ for validation (Line 1), ensuring main-task preservation while enabling adaptive poisoning calibration. The trigger δ is generated via Multi-dimensional Reinforcement Learning (MRL) (Line 5), enabling collaborative trigger evolution across malicious clients. Crucially, the

Algorithm 2 Pseudo Code of Our Backdoor Approach

Require: Global model Θ^t in round t , local dataset $\mathcal{D}_i$, reserved data fraction ρ_i , search granularity ζ , target class y^{tc} , reward function R , hyperparameter λ

Ensure: Local model update g_i^t

```

1:  $\mathcal{D}_i^{rev} \leftarrow \text{RandomSampling}(\mathcal{D}_i, \rho_i)$ ;  $\triangleright$  Reserve validation data
2:  $\text{count} \leftarrow \lceil 1/\zeta \rceil$ ;  $\triangleright$  Determine number of grid points
3:  $\text{max\_value} \leftarrow 0, o \leftarrow 0$ ;  $\triangleright$  Initialize best value and index
4: if  $i == \text{FirstIdOfAttackers}$  then
5:    $\delta \leftarrow \text{MRL}(\text{StateOfAttackers}, R)$ ;  $\triangleright$  get backdoor trigger through MRL
6: end if
7: for  $j = 0$  to  $\text{count}$  do
8:    $r \leftarrow \min(j \times \zeta, 1)$ ;  $\triangleright$  Candidate poisoning ratio
9:    $\mathcal{D}_i' \leftarrow \text{CreateBackdooredData}(\mathcal{D}_i \setminus \mathcal{D}_i^{rev}, \delta, y^{tc}, r)$ ;  $\triangleright$  Generate poisoned data
10:   $\tilde{\Theta}_{ij}^t \leftarrow \text{TrainingLocalModel}(\Theta^t, \mathcal{D}_i' \cup (\mathcal{D}_i \setminus \mathcal{D}_i^{rev}))$ ;  $\triangleright$  Train local model
11:   $\tilde{p}_{ij}^t \leftarrow 1 - \frac{1}{|\mathcal{D}_i^{rev}|} \sum_{x \in \mathcal{D}_i^{rev}} \mathbb{I}(G(x \oplus \delta \tilde{\Theta}_{ij}^t) = y^{tc})$ ;  $\triangleright$  Estimate genuine score
12:  if  $\tilde{p}_{ij}^t + \gamma r > \text{max\_value}$  then
13:     $o \leftarrow j$ ;  $\triangleright$  Update best index
14:     $\text{max\_value} \leftarrow \tilde{p}_{ij}^t + \gamma r$ ;  $\triangleright$  Update best value
15:  end if
16: end for
17:  $r_i^t \leftarrow \min(o \times \zeta, 1)$ ;  $\triangleright$  Optimal poisoning ratio
18:  $\mathcal{D}_i' \leftarrow \text{CreateBackdooredData}(\mathcal{D}_i, \delta, y^{tc}, r_i^t)$ ;  $\triangleright$  Generate final poisoned data
19:  $\Theta_i^t \leftarrow \text{TrainingLocalModel}(\Theta^t, \mathcal{D}_i' \cup \mathcal{D}_i)$ ;  $\triangleright$  Train final local model
20: return  $g_i^t = \Theta_i^t - \Theta^t$ .  $\triangleright$  Return model update

```

algorithm performs grid search over poisoning ratios $r \in [0, 1]$ with granularity ζ (Lines 7-16), evaluating candidate models $\tilde{\Theta}_{ij}^t$ through dual objectives. Stealth metric $\tilde{p}_{ij}^t$ measures evasion capability and Attack intensity r controls poisoning ratio. The Pareto-optimal solution maximizes $\tilde{p}_{ij}^t$ (Line 12-15), where γ regulates stealth-effectiveness tradeoffs. The optimal ratio r_i^t generates poisoned data (Line 18), training Θ_i^t (Line 19) to return malicious update g_i^t (Line 20). This optimization ensures persistent backdoor implantation while maintaining main-task fidelity through reserved data validation, demonstrating the necessity of protecting vulnerable populations from the threat attack to ensure fairness in the whole financial system.

D Pseudo Code of Our Orchestrated Backdoor Defense Approach

Our federated defense framework (Algorithm 3) implements a multi-stage suppression pipeline through coordination between client and server, achieving security and fairness in financial transactions. The algorithm initiates with parallel client-side computation where each participant k calculates local updates g_k^t (Line 4) and feature importance metrics $\mathcal{I}_k$ (Line 5) using Fisher information to quantify parameter sensitivity. Crucially, the server executes

Algorithm 3 Pseudo Code of Our Orchestrated Backdoor Defense Approach

```

1: for  $t = 1$  to  $T$  do
2:   Participant Side:
3:   for  $k = 1$  to  $K$  in parallel do
4:      $g_k^t \leftarrow \text{LocalUpdating}(\Theta^t, \mathcal{D}_k)$ ;  $\triangleright$  Optimize on private data
5:      $\mathcal{I}_k \leftarrow \text{ComputeImportance}(\Theta_k^t, \mathcal{D}_k)$ ;
6:   end for
7:   Server Side:
8:    $K_{clean} \leftarrow \text{Clean-Client-Selection}(\{w_k^t\}_{k=1}^K, \{\mathcal{I}_k\}_{k=1}^K, \Theta^t, \rho)$ ;
9:    $\Theta_a^t \leftarrow \Theta^t + \frac{1}{|S|} \sum_{i \in S} g_i^t$ ;  $\triangleright$  Construct auxiliary global model
10:   $\delta_{re} \leftarrow \text{REVERSEENGINEER}(\Theta_a^t)$ ;  $\triangleright$  Reverse engineer trigger
11:  for  $i \in K$  do
12:     $\Theta_i^t \leftarrow \Theta^t + g_i^t$ ;  $\triangleright$  Compute local model
13:     $C_{i,k}^t = \sum_{x \in \mathcal{D}_k} \mathbb{I}(G(x \oplus \delta_{re} \Theta_i^t) = y^{tc})$ ,  $k \in K_{clean}$ ;
14:     $p_i^t = 1 - \sum_{k \in K_{clean}} \frac{|\mathcal{D}_k|}{\sum_{k \in K_{clean}} |\mathcal{D}_k|} C_{i,k}^t$ ;  $\triangleright$  Compute genuine score
15:  end for
16:   $\Theta^{t+1} \leftarrow \Theta^t + \eta \cdot \frac{1}{\sum_{i \in S} p_i^t} \sum_{i \in S} p_i^t g_i^t$ ;  $\triangleright$  Update global model
17:  end for
18:   $\text{Clean-Client-Selection}(\{w_k^t\}, \{\mathcal{I}_k\}, w^t)$ 
19:  for  $k = 1$  to  $K$  in parallel do
20:     $g_k^t \leftarrow (\Theta_k^t - \Theta^t) / \eta$ ;  $\triangleright$  Compute gradient update
21:     $\tilde{g}_k^t \leftarrow g_k^t \odot \mathcal{I}_k$ ;  $\triangleright$  Reweight with importance
22:  end for
23:   $\tilde{g}^t \leftarrow \sum_k \alpha_k \tilde{g}_k^t$ ;  $\triangleright$  Aggregate global gradient
24:   $V \leftarrow \text{ComputeGradientVariance}(\tilde{g}^t, \{\tilde{g}_k^t\})$ ;
25:   $K_{clean} \leftarrow \text{SORTED}(V, \rho)$ ;
26: return  $K_{clean}$ .

```

clean-client-selection (Line 8) by reweighting gradients $\tilde{g}_k^t$ to amplify anomalies in trigger-sensitive parameters (Lines 19-20), then selecting the ρ portion of clients with minimal gradient variance V (Lines 22-24). This Fisher-driven screening identifies high-fidelity participants while filtering potential malicious actors in financial scenarios.

The core defense mechanism constructs an auxiliary global model Θ_a^t (Line 9) intentionally amplifying backdoor susceptibility through uniform aggregation. This enables reverse trigger engineering δ_{re} (Line 10) via optimization. For each client i , the defense computes Local model Θ_i^t (Line 12) Trigger susceptibility $C_{i,k}^t$ (Line 13) of all clients evaluated on clean client k Credibility score p_i^t quantifying evasion capability (Line 14).

The global model update (Line 16) employs adaptive aggregation dynamically to suppress malicious updates through credibility-weighted averaging in inclusive finance. This closed-loop system achieves real-time backdoor mitigation by combining gradient variance analysis for client screening, auxiliary model construction for trigger inversion, and susceptibility-based reweighting for poisoned update suppression to provide underserved communities with reliable protection from backdoor threats to guarantee safety and fairness in web inclusive finance.